TAN TAO UNIVERSITY
SCHOOL OF INFORMATION TECHNOLOGY

TAN TAO UNIVERSITY
FROM KNOWLEDGE TO THE STARS

Final Project: Customer Lifetime Value & Revenue Loss from Churn

Course: Data Visualization (CS441V)

Spring 2026

Group: 1

Members	**Student Code**
Trieu Minh Thuan	2302059
Ho Nguyen Kim Long	2102161

Supervisor: Nguyen Xuan Ha

Table of Contents

1. Introduction

1.1 The financial lens on churn

Customer churn is conventionally measured as a percentage — "we lose 26.5% of customers per period". That number is easy to track but hides the question every executive actually wants answered: how much money are we losing? A 26.5% churn rate means very different things if the customers we lose are mostly low-value newcomers (a small revenue dent) versus mostly long-tenured premium accounts (a major financial event).

For this Final Project we replace the descriptive lens of the Midterm with a financial one. We compute Customer Lifetime Value (CLV) for every customer, identify who creates that value, quantify the dollars destroyed by churn, and translate the result into an explicit Return-on-Investment case for retention spending. The same dataset, but a sharper question.

1.2 Core question and five sub-questions

Core question: *How much is each customer worth, who creates value, and how much money is the company losing to churn?*

#	Sub-question
Q1	How is Customer Lifetime Value distributed? Does the 80/20 rule hold here?
Q2	What does a top-20% “Gold” customer look like, vs a bottom-20% “Bronze” one?
Q3	How much revenue does churn actually destroy — past and future?
Q4	How does CLV grow with tenure (cohort analysis)?
Q5	What is the ROI of retention spending — at 10%, 20%, 30% save rates?

2. Data and CLV Definition

2.1 Dataset

The IBM Telco Customer Churn dataset (Kaggle / IBM Sample Data Sets) contains 7,043 customer records and 21 columns covering demographics, contract details, services subscribed to, billing information, and the churn label. We use TotalCharges as the proxy for cumulative revenue per customer, and MonthlyCharges as the run-rate.

2.2 Defining Customer Lifetime Value

CLV is decomposed into two components, separating revenue already received from revenue at risk:

- **CLV_realized** = TotalCharges already paid (past revenue, no longer at risk).
- **CLV_future** = expected remaining revenue over a 3-year horizon. For staying customers: MonthlyCharges × max(0, 36 − tenure). For churners: 0.
- **CLV_lost** = the future revenue that would have been received from churners had they stayed. Calculated identically to CLV_future but only for the 1,869 churned customers.

This is a deliberately simple model — no discount rate, no cost-of-service. The goal is interpretability for stakeholders, not actuarial precision.

2.3 Improvement vs the Midterm

The Midterm Project standardized numeric features with StandardScaler before plotting, so axes showed z-scores between -2 and +2 — uninterpretable for a business reader. This Final Project keeps months and US dollars throughout. Charts can be read directly without reverse-engineering a scaler.

3. Q1 — How is CLV distributed?

The first question for any value-based analysis is whether revenue is concentrated in a few customers (Pareto pattern) or spread evenly. The answer determines whether to focus retention effort on a small number of high-value accounts, or to invest in broad mass-market programs.

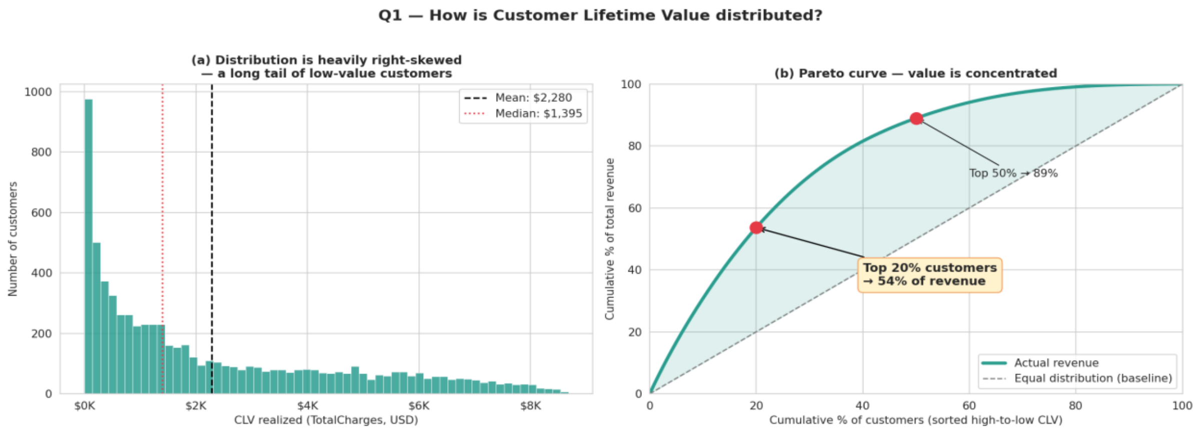

Figure 1. (Left) CLV distribution is heavily right-skewed — mean is 63% above median. (Right) Pareto curve with 20% and 50% milestones annotated.

The 80/20 rule does not hold exactly, but a milder version of concentration does:

Top % of customers	Cumulative % of revenue
Top 10%	31%
Top 20%	54%
Top 30%	70%
Top 50%	89%

A textbook Pareto would say "top 20% generate 80% of revenue". Here it is 54% — still highly concentrated but flatter than a strict Pareto. The shape: a heavy right tail of low-CLV customers (mostly newcomers who have not yet accumulated much), and a thin spike of high-CLV customers (long-tenured premium veterans). The mean CLV ($2,280) is 63% higher than the median ($1,395), confirming the skew.

→ **Action:** Top 20% holds half the book, but the long tail still matters. Retention effort should be tiered, not concentrated solely at the top.

4. Q2 — Profile of Gold vs Bronze customers

We split customers into 5 quintiles by CLV_realized and compare the top quintile (Gold, top 20%) with the bottom quintile (Bronze, bottom 20%) to identify what drives high lifetime value.

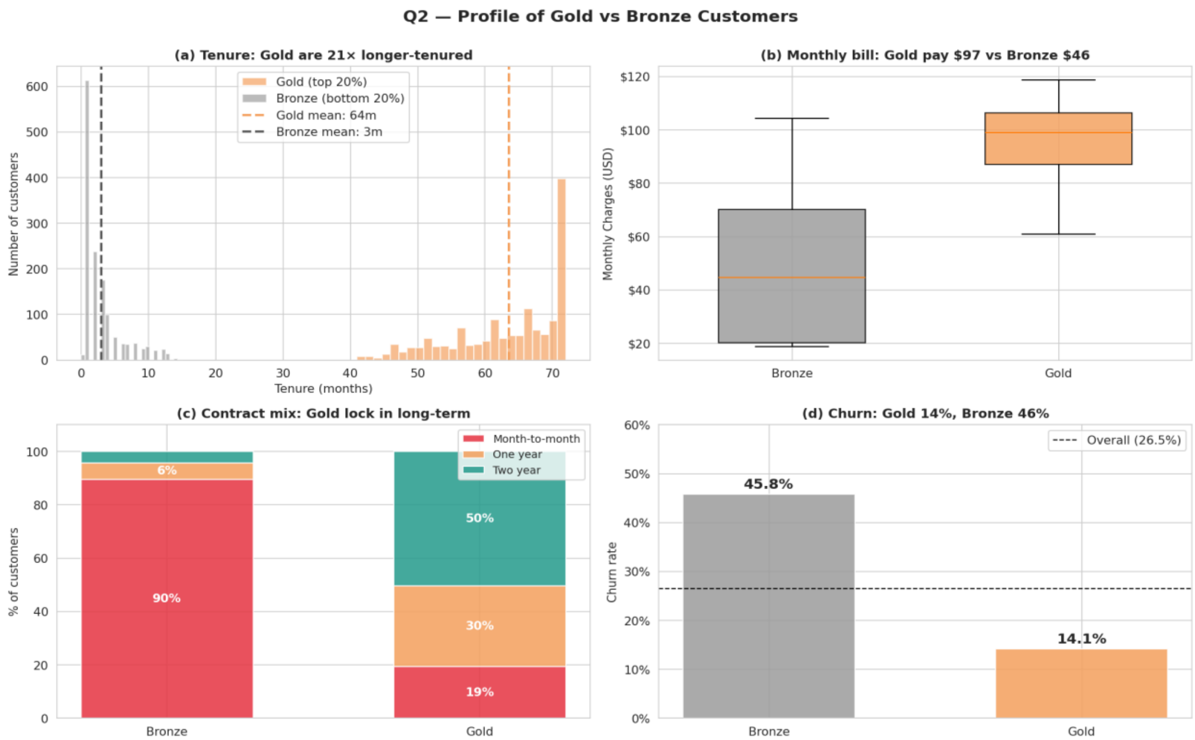

Figure 2. Profile comparison across four dimensions: tenure distribution, monthly charges, contract mix, and churn rate.

	Bronze (bottom 20%)	Gold (top 20%)
Avg tenure	3 months	63 months
Avg monthly charges	$46	$97
Avg CLV realized	$115	$5,820
Churn rate	45.8%	14.1%
% Two-year contract	8%	51%
% Auto-pay	23%	60%

Gold customers are not just slightly better. Their average past revenue is 51× that of Bronze ($5,820 vs $115). Even at a 14% churn rate, losing one Gold is more painful than losing 3–4 Bronze. The math of retention is non-linear in customer value.

The Gold formula: long tenure + premium plan + long-term contract + auto-pay. The Bronze profile is the inverse on every one of those dimensions. This gives us a clear picture of what to nudge a Bronze customer toward in order to convert them into a Gold trajectory.

5. Q3 — The financial damage from churn

We connect CLV to churn directly. We compare CLV between churners and stayers, then quantify the future revenue forfeited when each churner left.

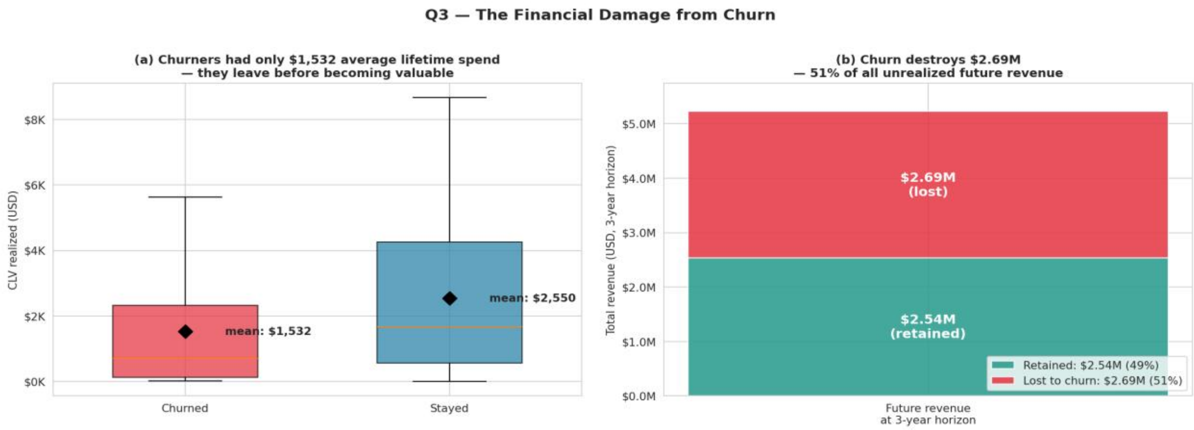

Figure 3. (Left) Boxplot of CLV realized by churn status, with means marked as black diamonds. (Right) Stacked bar showing future revenue split into retained vs lost-to-churn.

- **Churners die young.** Average lifetime spend is $1,532 vs $2,550 for stayers — a 40% gap. The gap exists mostly because churners did not survive long enough to accumulate value.
- **The big number is the future.** Future revenue lost from already-churned customers is $2.69 million over the 3-year horizon. Compared to the $2.54M of future revenue that staying customers will deliver, this means 51% of all future revenue that could have been collected is forfeited.
- **Per-churner cost: $1,440.** On average, every churner costs the company $1,440 in lost future revenue — a useful benchmark for sizing retention budgets.

The phrase "we lose 26.5% of customers" hides the real story. Those 26.5% were on track to deliver $2.69M more if retained. That number is what any retention investment is competing against.

6. Q4 — CLV Cohort Analysis

CLV does not appear overnight — it accumulates month by month. We bin customers by tenure and watch average realized CLV grow alongside churn rate dropping.

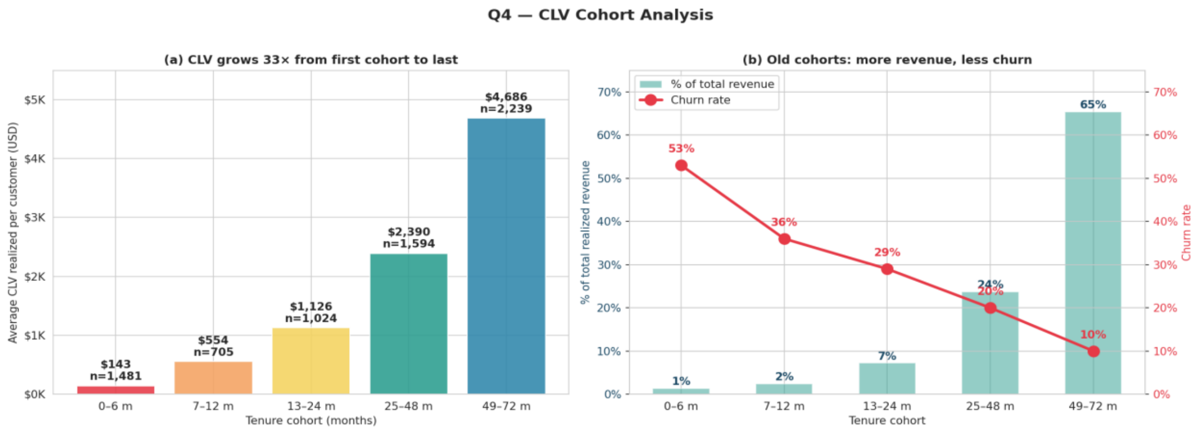

Figure 4. (Left) Average CLV per customer rises 33× from the youngest cohort to the oldest. (Right) Dual-axis: revenue contribution (bars) climbs from 1% to 65% while churn rate (line) drops from 53% to 10%.

Cohort	Avg CLV	Churn rate	% of total revenue
0–6 m	$143	53%	1%
7–12 m	$554	36%	2%
13–24 m	$1,126	29%	7%
25–48 m	$2,390	20%	24%
49–72 m	$4,686	10%	65%

Two complementary curves: as cohorts mature, CLV grows 33× from first to last, while churn drops 5×. The oldest cohort generates 65% of all realized revenue with the lowest churn rate. The youngest cohort is simultaneously the lowest-revenue and the highest-risk group — a dangerous combination, but also the highest-leverage place to intervene.

> → **Action:** Every customer who survives the early months stops being a cost and starts being a profit center. Retention investment in early-tenure customers compounds: we do not just save one customer-year of revenue, we save their entire growth trajectory toward the high-CLV cohorts.

7. Q5 — ROI of retention spending

We turn the analysis into an explicit financial decision. Suppose we run a retention campaign costing $50 per targeted customer (industry-typical), and suppose we save a fraction of those targeted. What is the expected payback?

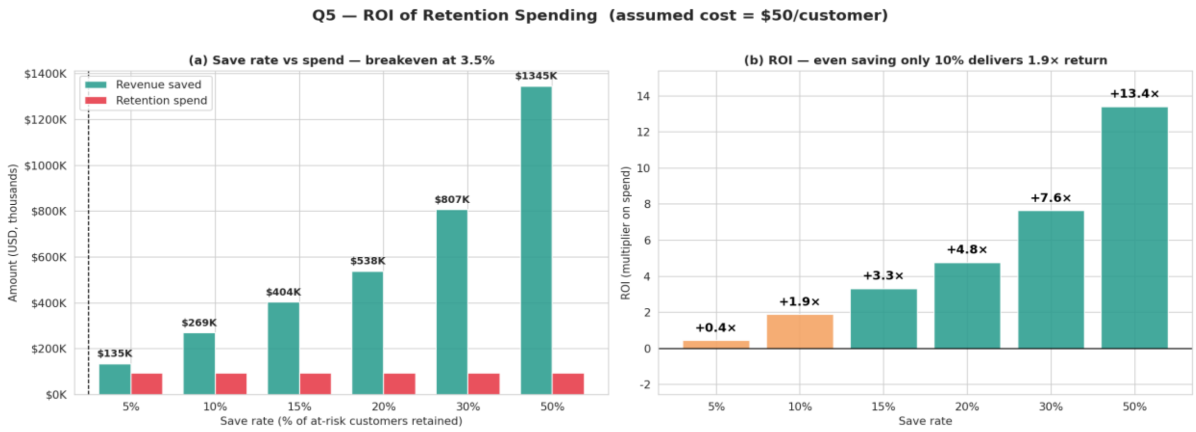

Figure 5. (Left) Revenue saved versus campaign spend at each save rate. Vertical dashed line marks the breakeven point at 3.5%. (Right) ROI multiplier per save rate — even a modest 10% save rate yields 1.9× return.

Save rate	Revenue saved	Net profit	ROI
5%	$135K	$41K	0.4×
10%	$269K	$176K	1.9×
15%	$404K	$310K	3.3×
20%	$538K	$445K	4.8×
30%	$807K	$714K	7.6×
50%	$1,345K	$1,252K	13.4×

The breakeven save rate is just 3.5% — saving one in 28 at-risk customers is enough for the campaign to pay for itself. Industry norms for telecom retention range from 10% to 30%, putting expected ROI between 2× and 8×. That is the kind of return that turns a retention program from a discretionary expense into a no-brainer.

Caveats. The calculation assumes perfect targeting (we know in advance which 1,869 customers will churn). Real campaigns rely on imperfect risk scores, so effective ROI is somewhat lower than the headline numbers. Even with a 50% targeting accuracy, however, the breakeven save rate stays under 10% — still well below typical campaign performance.

8. Comparison with the Midterm Project

The Midterm asked one broad question and answered it with three KDE plots showing univariate distributions of tenure, monthly charges, and total charges split by churn status. This Final Project re-frames the same dataset through a financial lens.

Aspect	Midterm	Final Project
Lens	Statistical (% churn)	Financial (dollar CLV)
Question scope	One broad question, surface-level	One financial problem with 5 layered questions
Visualizations	3 KDE plots (univariate)	5 multivariate financial figures
Chart types	KDE, bar	Histogram + Pareto, profile multi-panel, waterfall, dual-axis cohort, ROI scenarios
Pre-processing	Standardized before plotting (axes meaningless)	Raw dollars and months for interpretability
Key metric	Churn rate (%)	CLV ($), revenue at risk ($), ROI (multiplier)
Decision support	"Customers with low tenure churn more"	"Save 10% of churners → $176K net profit, ROI 1.9×"

A concrete example of depth

The Midterm tenure plot showed: *"Churned customers tend to have lower tenure."* True, but uninformative on its own.

This Final Project translates the same observation through a financial lens: Bronze customers (bottom 20% by CLV) churn at 46% but each only represents $115 of past revenue — losing one costs little. Gold customers (top 20%) churn at 14% but each represents $5,820 of past revenue plus an unknown future. The arithmetic of retention is non-linear in customer value, and CLV — not churn rate — is the right metric to optimize. That single re-framing turns the analysis from descriptive into prescriptive.

9. Conclusions

9.1 Key insights

- **Revenue is concentrated, but not extreme.** Top 20% of customers generate 54% of revenue. The 80/20 rule is softer than the textbook version, but priorities should still be tier-aware.
- **Gold customers are a different species.** 63 months tenure, $97 monthly charges, two-year contracts, auto-pay, 14% churn. Bronze customers are the inverse on every dimension.
- **Churn destroys $2.69M in future revenue** over a 3-year horizon — about half of all future revenue that could otherwise have been collected.
- **CLV is a lifecycle.** It grows 33× from the first cohort to the last, while churn drops from 53% to 10%. The oldest cohort delivers 65% of revenue with the lowest churn — value compounds for every customer who survives early tenure.
- **Retention spending pays off easily.** Breakeven is just 3.5% save rate. Even a modest 10% save rate yields 1.9× ROI; a typical 20–30% save rate yields 5–8×.

9.2 Recommended retention playbook

Priority	Target	Action	Why
1	New customers (0–12 m) on premium plans	Onboarding bundle: tech-support trial, contract upgrade discount	High Bronze→Gold conversion potential; saving them captures a 33× CLV trajectory
2	Existing Gold-quintile customers showing risk signals	Personal account manager, loyalty bonus	Avg CLV $5,820 — saving one Gold equals saving 50 Bronze
3	Mid-tenure customers (13–24 m)	Contract renewal incentives	About to enter the most profitable phase; ROI of saving them is highest
4	Low-CLV high-monthly-charges customers	Plan downgrade options to reduce churn	Better to keep at lower revenue than lose entirely

9.3 Limitations

Static CLV model. We use point estimates, not a probabilistic stream — no discount rate, no cost-of-service, no churn probability decay over time. A more rigorous model would treat each customer's future as a probability-weighted cash flow.

Three-year horizon is arbitrary. A longer horizon would show even more revenue at risk; a shorter one would compress the picture. Three years was chosen because it matches the longest contract length in the dataset and is the typical planning horizon for telecom executives.

Idealized retention targeting. We assume we know in advance who the 1,869 churners will be. Real campaigns rely on risk scores with imperfect precision, so effective ROI in production is somewhat lower than the headline numbers above.

Correlation, not causation. CLV and churn correlate with contract type, payment method, and tenure, but we cannot prove that intervening on those features will cause the same lifetime extension we observe in stayers. Validating the recommended actions would require a controlled A/B test.

Author Contributions

Member	Student Code	Contribution	%
Trieu Minh Thuan	2302059	CLV mathematical formulation, Q1 (distribution and Pareto), Q2 (Gold vs Bronze profile), Q5 (ROI scenarios); report writing for Sections 1, 3, 4, 7–9.	**50%**
Ho Nguyen Kim Long	2102161	Data preparation pipeline, Q3 (revenue lost from churn), Q4 (cohort analysis); report writing for Sections 2, 5, 6.	**50%**